

Test for the mean. Population variance known

Data scientist salary

Dataset			
\$ 117,313	Sample mean	\$ 100,200	H_0 (Glassdoor data) \$ 113,000
\$ 104,002	Population std	\$ 15,000	Two-sided test
\$ 113,038	Standard error	\$ 2,739	
\$ 101,936			
\$ 84,560	Sample size	30	
\$ 113,136			
\$ 80,740	Two-sided test		
\$ 100,536	$Z_{0.025}$	1.96	
\$ 105,052	$Z_{0.005}$	2.58	
\$ 87,201			
\$ 91,986	Z-score	-4.67	
\$ 94,868			
\$ 90,745			
\$ 102,848			
\$ 85,927			
\$ 112,276			
\$ 108,637			
\$ 96,818			
\$ 92,307			
\$ 114,564			
\$ 109,714			
\$ 108,833			
\$ 115,295			
\$ 89,279			
\$ 81,720			
\$ 89,344			
\$ 114,426			
\$ 90,410			
\$ 95,118			
\$ 113,382			

If your z-score is **below** the critical value (not in the rejection region), it means you didn't reach the line. You are still in the "safe zone" or the middle of the crowd where things are considered normal.

When your z-score is below the critical value, we say that you **fail to reject the null hypothesis** (H_0)

The **null hypothesis** is the status quo and any differences are just down to luck or chance. Because the score didn't cross the boundary, there's not enough evidence to say that anything unusual is going on (the result is not statistically significant).